

Detection and Defense from False Data Injection Attacks In Aviation Cyber-Physical Systems Using Artificial Immune Systems

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Abstract—In recent years, there has been a rapid expansion in the development of Cyber-Physical Systems (CPS), which allows the physical components and the cyber components of a system to be fully integrated and interacted with each other and with the physical world. The commercial aviation industry is shifting towards Aviation Cyber-Physical Systems (ACPS) framework because it allows real-time monitoring and diagnostics, real-time data analytics, and the use of Artificial Intelligent technologies in decision making. Inevitably, ACPS is not immune to cyber-attacks due to integrating a network system, which introduces serious security threats. False Data Injection (FDI) attack is widely used against CPS. It is a serious threat to the integrity of the connected physical components. In this paper, we propose a novel security algorithm for detecting FDI attacks in the communication network of ACPS using Artificial Immune System (AIS). The algorithm was developed based on the negative selection approach. The negative selection algorithm is used to detect malicious network packets and drop them. Then, a Nonlinear Autoregressive Exogenous (NARX) network is used to predict packets that dropped by the negative selection algorithm. The developed algorithm was implemented and tested on a networked control system of commercial aircraft as an Aviation Cyber-physical system.

Keywords—AIS, NARX, CPS, ACPS, Immune System, Negative Selection, TCM, Cyberattacks, FDI Attacks

I. INTRODUCTION

Cyber-Physical Systems (CPS) is an intelligent system that enables real-time interaction between the physical components of a system using cyberspace [1]. It is a form of the Industrial Internet of Things (IIoT) and is considered as part of the fourth industrial revolution (industry 4.0), which is a natural evolution of IoT [2]. CPS allows the integration of physical processes, computation, communication, and control systems [1]. CPS development provisions many benefits for the industry, such as cost reduction, increasing the system capability, and performance [3]. The aviation industries recently showed an interest in transferring their aircraft into fully connected and network aircraft systems [8]. This desire will transform aircraft in general, and commercial aircraft in particular into Aviation Cyber-Physical Systems (ACPS). Using the ACPS framework allows real-time monitoring and pilots to avoid collisions early, especially

when two or more aircraft become closer to the same zone [4].

However, ACPS, like any other CPS, is susceptible to security threats that could impact the operation of the physical components due to the possibility of conducting cyber-attacks through the communication network [5]. Therefore, the security of ACPS is one of the aviation industry's top priorities to ensure the safety of air transportations [6]. It is a critical and challenging issue due to the integration of the network, which increases the level of threats against the physical components of aircraft. Cyber-attacks pose risks to the ACPS similarly to the risk they cause to traditional IT platforms, i.e., cloud computing, leading to catastrophic consequences [6]. Researchers are making efforts to enhance the security of ACPS, but it is not sufficient yet [7]. It was reported that cyber-attacks were responsible for the two recent Boeing 737 Max 8 crashes [8]. It is believed that the controlled sensors data of the Maneuvering Characteristics Augmentation System (MCAS) received false readings in both commercial aircraft. Unauthorized access through the network system of the ACPS could result in mechanical failures or malfunctioning of the system's components since attackers can inject false data into the physical components of the aircraft. This type of attack is known as Fault Data Injection (FDI) attacks [5]. FDI attack is a cyber-attack that is widely used against CPS [6]. It considers a serious threat to the integrity of the connected physical system's components because adversaries can inject false data into those connected devices such as sensors and actuators [9]. Generally, it can be challenging to detect FDI attacks because of the attack's behavior, and their impacts might not be immediately observed [10].

This paper proposes a novel security method and algorithm for detecting FDI attacks in the communication network of ACPS using Artificial Immune System (AIS). The algorithm is based on a negative selection approach that exists in the natural immune system. In other words, the negative selection algorithm is developed to identify malicious network packets to drop them. Then a Nonlinear Autoregressive Exogenous (NARX) network is used to forecast packets that were dropped by the negative selection algorithm.

The rest of the paper is organized as follows: Section II represents the related research works. In Section III, we present and discuss the system architecture. The proposed research methodology is explained in section IV. In section V the real-time simulation and test results are presented and discussed. Finally, section VI provides the conclusion of the research work.

II. RELATED WORK

In this section, we review the recent research works that have been reported on detecting and defending CPS from FDI cyber-attacks.

Due to the catastrophic consequences of FDI attacks on the CPS, researchers proposed various approaches to mitigate their impacts. In the research work reported in [11], the authors proposed an adaptive neural network to detect FDI attacks in the UAV system by classifying fault data from the true data. In the work, two separate FDI attacks were injected into the Inertial Measurement Units (IMU). The IMU is a sensor that is used to measure the angular and linear motion of the aircraft (UAV). The first attack was the step-shape attack, and the second attack was the Gaussian-shape attack. It was also mentioned that conventional NN is not sufficient to be used due to its slow learning rate. Therefore, the authors used Embedded Kalman Filter (EKF) to update the learning coefficients, which improved the detection results [11].

The authors in [12] implemented a method to detect and mitigate FDI attacks on a car-following system using Interval type-2 fuzzy logic systems (IT2- FLS). Two vehicles were used in the experiment, the leader vehicle, and the follower vehicle. Typically, the follower vehicle should follow the leader vehicle. To simulate the attacking scenario, a fault velocity was injected into the nonlinear acceleration model of the follower vehicle; therefore, the follower vehicle lost track of the leader vehicle. However, in the second scenario, the IT2- FLS was used to design an interval type-2 fuzzy PID controller to mitigate the cyber-attack and handle uncertainty. According to the authors, the interval type-2 fuzzy PID controller provided better performance than the classical PID-controller. Although it is easy to design a PID controller using a fuzzy logic approach, it requires a high level of knowledge [13, 14]

In the research reported in [10], the authors studied the effect of FDI attacks on the Predictive Maintenance (PdM) that was used with the aircraft system. PdM is a method used to analyze production data to prevent the failure of in-service equipment such as a turbofan engine by estimating issues before they occur. Therefore, PdM is considered an industry 4.0 solution to estimate the Remaining Useful Life (RUL) of in-service equipment. The authors used PdM to predict the health of the turbofan engine using a dataset that was provided by NASA's C-MAPSS (Commercial Modular Aero-Propulsion System Simulation). The goal of the attacker was to modify the sensors reading that transmits to the PdM system. Three deep learning algorithms were involved in the

experiments, which are Convolutional Neural Network (CNN), Gated Recurrent Unit (GRU), and Short-Term Memory (LSTM). Those three algorithms were evaluated to see which one produces tremendous results among each other. According to the authors, GRU obtained better results. The only limitation of this research is that the experiments were not conducted in real-time simulation. Although deep learning is very efficient to use, it is difficult to decide which deep learning rule is appropriate because there is no stander theory that can guide researchers [15].

III. SYSTEM ARCHITECTURE

This section represents the architecture of the ACPS that is used in this research.

Generally, CPS comprises five subsystems: the physical process (plant), controllers, sensors, actuators, and networks, as shown in Fig. 1.

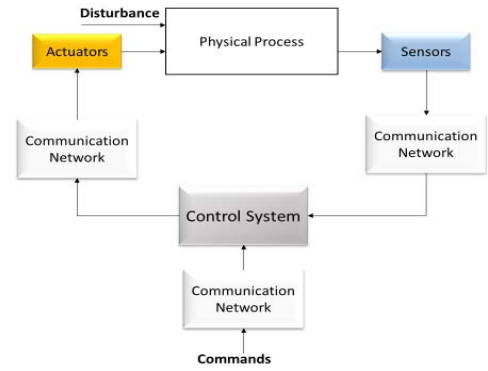


Fig. 1. General Architecture of CPS

A. Physical Process

The physical system used in this research is a commercial aircraft system that is called Transport Class Model (TCM). The TCM is a full-scale commercial aircraft model developed by NASA for the research community [16]. It simulates the dynamics of commercial aircraft using MATLAB/Simulink. The original TCM, developed by NASA, is a classical system model that receives inputs, processes them, and then produces outputs, as illustrated in Fig. 2. Therefore, the original TCM model does not include any controller or communication network involved between its different components.

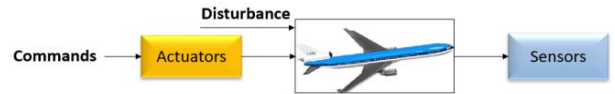


Fig. 2. The Original TCM Model

B. Control System

The control system (developed in previous work of the authors) used in this research is the Model Reference Adaptive Control (MRAC), which is an adaptive controller integrated

with the aircraft model. It is an advanced intelligent control system that can ensure the output of the controlled system (aircraft) to follow the output of a reference model [17]. MRAC system uses an adaptive control strategy to guarantee closed-loop stability [18]. The development and implementation of MRAC were reported in [17]. The control system was successfully integrated into the TCM.

C. Network System

The Network system was developed, by the authors, using SimEvents to simulate a real-time network system based on a discrete-event approach. SimEvents is a MATLAB/Simulink library that provides some features such as monitoring network traffic by measuring essential network traffic parameters, such as server utilization, departure packets, and blocked packets [19]. SimEvents also is a great tool that can be used for network traffic analysis.

D. Establishment of ACPS

To establish the ACPS system, the developed control system, and the communication networks were integrated to transform the aircraft model (TCM) into an aviation cyber-physical system (ACPS), as illustrated in Fig. 3. Hence, all sensors and actuators of the aircraft will be communicating and exchanging data and information through the added communication network. Sensors detecting and sensing responses of the aircraft will transmit its data to the control system through the added communication network. Also, the controllers generating control signals and commands to the flight surfaces of the aircraft will be transmitted through the same communication network.

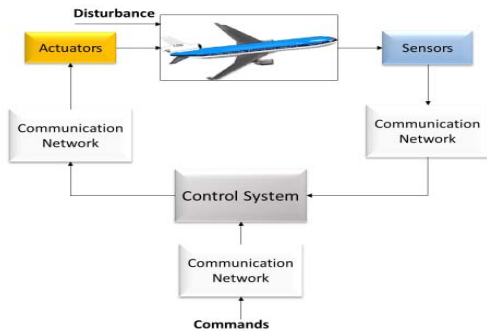


Fig. 3. High-level Depiction of ACPS Architecture

IV. RESEARCH METHODOLOGY

A. Detection Method Using AIS

This section discusses the detection method used in this research using Artificial Immune Systems (AIS).

AIS is a machine learning algorithm inspired by the Natural Immune Systems (NIS) of human beings. Therefore, NIS's theory is used in AIS to model non-biological applications [20, 21]. There are several AIS models used in previous research, such as:

- Classical View Models
- Clonal Selection Theory Models

- Network Theory Models
- Danger Theory Models

In our research, we are only focused on the Classical View Models, precisely the negative selection approach.

A negative selection algorithm is used to train T-cells to distinguish between self and non-self (antigen) cells. T-cell is a lymphocyte cell that is created in the bone marrow of the human body [20]. Therefore it can provide a very accurate detection mechanism to detect malicious network packets [22]. The first attempt to design a negative selection algorithm was by Forrest, in which a simple Artificial Lymphocytes (ALC) was trained [22].

1) Training Procedure

The algorithm shown in Fig. 4 (Algorithm-1) was used to train the ALC cells using a negative selection approach. In step-1, the true_input (original elevator signal) and the input (false elevator signal) are generated. In steps, 2-3, the start and end of the training are identified. In step-4, the ALC matrix size is defined, and in step-5, ALC is generated randomly. The threshold is determined in step-7. The loop for training ALC cells starts from step-7 to step -20. The main goal of this loop is to distinguish between self ALC cells and non-self ALC cells (antigens).

Algorithm 1: Training of ALC cells using Negative Selection

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1: Generate true_input and input
2: Set the counter n which is the start of the training
3: Set the counter en which is end of training ALC
4: Set the size of the ALC matrix row , column
5: Randomly generate ALC (row, col)
6: Set the threshold
7: while n <= en
8:   for i = 1:row
9:     for j = 1:col
10:      if ALC(i,j) >= true_input(i) + threshold
11:        ALC(i,j) = ALC(i,j)
12:      else if ALC(i,j) <= true_input(i) - threshold
13:        ALC(i,j) = ALC(i,j)
14:      else
15:        ALC(i,j) = random(1,1)
16:      End if
17:    End for
18:  End for
19:  n = n + 1
20: End while

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Fig. 4. The Training ALC

2) Detection Procedure

Part-2 of the algorithm (Algorithm-2) shown in Fig. 5 was used to detect FDI attacks using the train ALC cells from algorithm-1. Step-1 calls the generated input, and step-2 defines the differ matrix to store the differences between input and the trained ALC cells. Step-3 defines the ALC_test matrix, which stores the small differences of the differ matrix. The loop for the detection procedure starts from step-7 to step-15, and its main goal is to detect malicious network packets based on the assigned threshold.

3) Testing the Detection

The detection method was tested to confirm its validity. Fig. 6 shows the original and false elevator signals together. It can be easily observed the similarity and dissimilarity between the two signals sent to the elevator of the aircraft. Fig. 7 illustrates the results of the detection algorithm. The black line indicates the time where the FDI attack was detected.

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Algorithm 2: Detecting of FDI attack
1: input           //use the same input generated from training step
2: Set the differ=0 //to store the differences between the input and the ALC
3: Set the ALC_test=0 //to store only the smallest differences
4: Set the size of the ALC matrix row , column
5: ALC (row, col) //use the trained ALC from the training step
6: set the threshold
7: for i=1:row
8:     for j=1: col
9:         differ (i,j)=abs(injected_input (i,1)-ALC(i,j))
10:    End for
11:    ALC_test(i)=min(differ(i,end)) //only a small differences is matter
12:    if ALC_test(i)<= threshold
13:        malicious packet is detected
14:    End if
15:End for

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Fig. 5. The Detection Algorithm

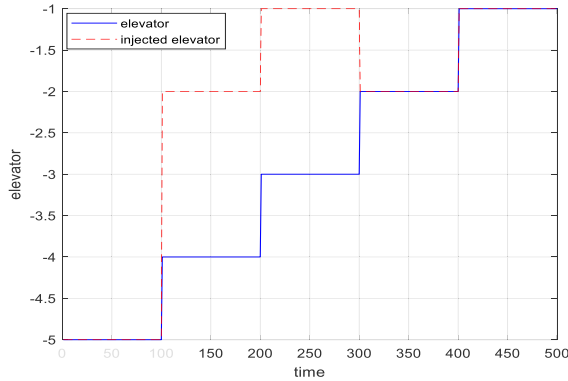


Fig. 6. Original and Injected Signals

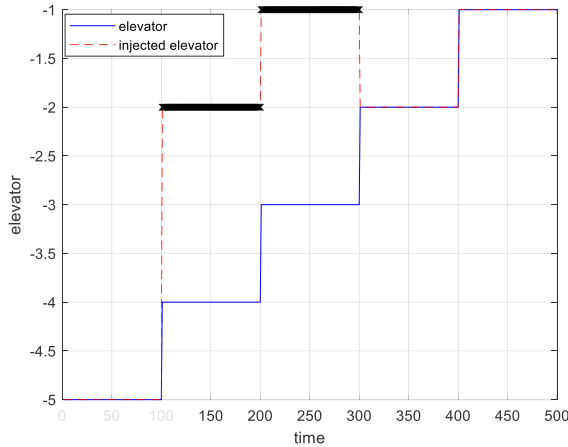


Fig. 7. Detection Result

B. Defending method using NARX

Nonlinear Autoregressive Exogenous (NARX) network is used to estimate network packets that were dropped by the detection mechanism. A Series-Parallel Architecture of NARX was used, and it composes the input layer, hidden layer, and output layer. The input layer consists of two nodes one receives input, and another gets the true output. The hidden layer comprises ten nodes, and the output layer has only one node that produces the predicted packet. The NARX network was trained using the Levenberg–Marquardt algorithm. The trained input is the false elevator signal. We used 500,001 input data points, 70% of the input data used for training, 15% for testing, and 15% for validation. The regression plot of the training, testing, and validation are shown in Fig. 8.

V. SIMULATION AND TEST RESULTS

The process of testing and simulation implemented in this research work is shown in Fig. 9. The FDI attack is injected with the pilot commands, which is the elevator command signal in our case. After the attack is injected, then the detection is activated using the AIS algorithm to identify and drop the malicious packets. Finally, the NARX network is used to predict the lost packets.

The initial conditions of the state variables of the aircraft are shown in Table 1. The state variable u represents the velocity in the x-direction, ft/sec, w refers to velocity in the z-direction, ft/sec, q body pitch rate, rad/sec, and Θ is Euler angle for pitch, rad. In addition, the Calibrated Airspeed (cas) is trimmed at 250 knots, and the altitude is trimmed at 10,000 ft. The simulation was performed when the aircraft was on the cruise flight.

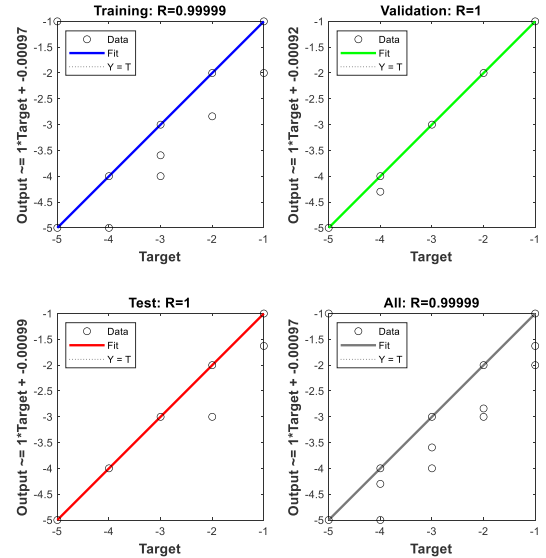


Fig. 8. Regression Plots Training, Validation, Test and All

Table 1 Aircraft Initial Conditions

State variables	Trim condition
u	485.69 ft/sec
w	38.72 ft/sec
q	0 rad/sec
θ	0.079 rad

We compared the aircraft response of the state variables u , w , and the angle of attack (AOA) under FDI attacks to the reference model signals. Also, we compared the aircraft responses of u , w , and AOA under FDI attack using the proposed detection and defense method to the signal provided by the reference model. The response of the speed of the aircraft " u " under an FDI attack is shown in Fig. 10. The red line represents the actual aircraft performance, and the blue line represents the reference model signal that the aircraft must follow. It is clear that the aircraft did not have a stable performance between 100 and 300 seconds due to the FDI attack. However, Fig. 11 shows the response of the aircraft speed " u " using the proposed detection and defense method. The aircraft speed follows the reference model speed. The performance of the aircraft state " w " is shown in Fig. 12. We can observe that the response of the aircraft " w " state variable is not stable, especially in the first 200 seconds. However, Fig. 13 illustrates the performance of the aircraft speed " w " using the proposed detection and defense method. Finally, Fig. 14 shows the performance of aircraft AOA under an FDI attack, and Fig. 15 shows its performance using the proposed detection and defense method. It can be observed that the proposed detection and defense method from FDI cyber-attacks has successfully returned the aircraft to its stable states in all the four test scenarios.

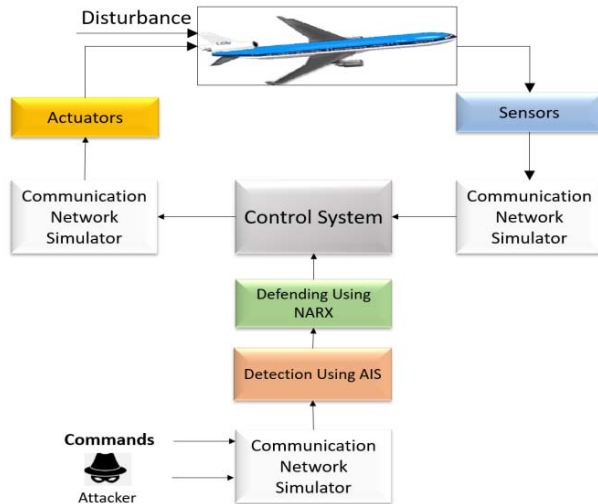


Fig. 9. Simulation Process

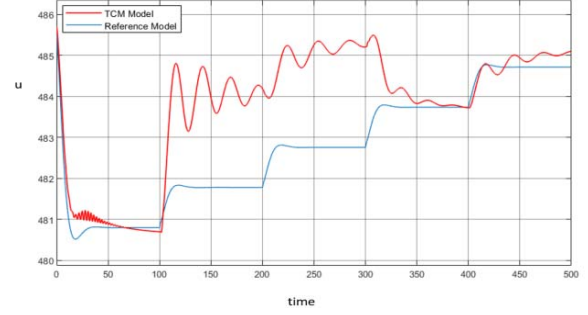


Fig. 10. Aircraft Speed " u " under FDI Attack

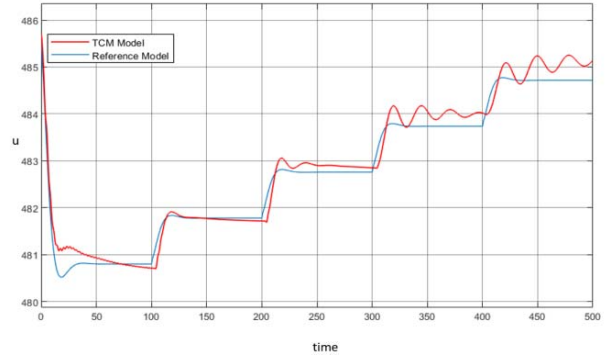


Fig. 11. Aircraft Speed " u " using the proposed Defense method

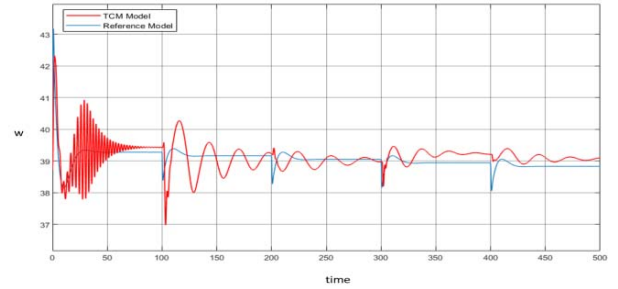


Fig. 12. Aircraft Velocity " w " under FDI Attack

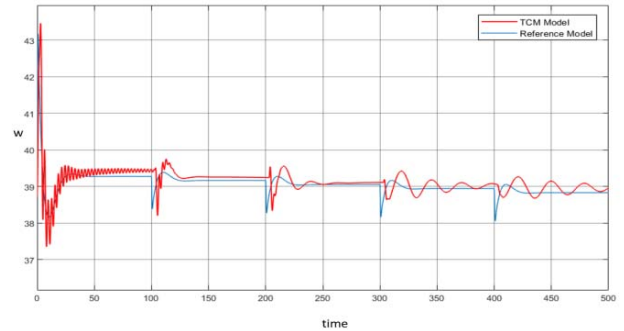


Fig. 13. Aircraft Velocity " w " using the proposed Defense method

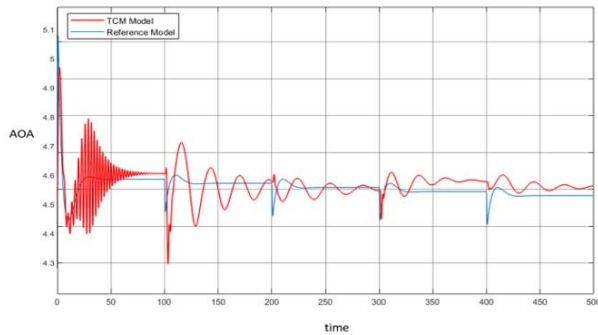


Fig. 14. AOA of the Aircraft under FDI Attack

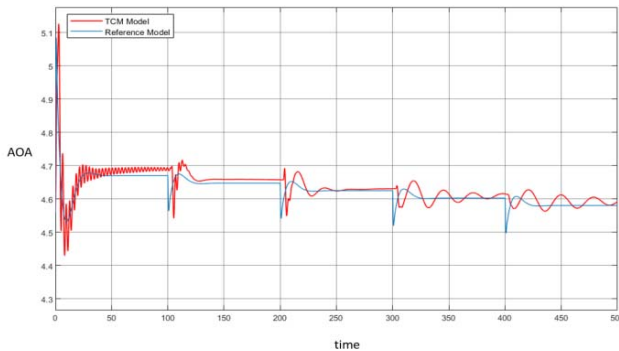


Fig. 15. AOA the Aircraft using the Proposed Detection and Defense Method

VI. CONCLUSIONS

In conclusion, a novel algorithm for detecting False Data Injection (FDI) type of cyber-attacks in Aviation CPS systems using Artificial Immune System (AIS) based on the negative selection approach was developed, implemented, and tested. The process of implementing and testing the proposed cyber-security algorithm was as followed. First, a commercial aircraft model TCM was converted into a CPS by integrating an adaptive control system with the aircraft model through a communication network. Second, malicious packets were injected into the control system through the network nodes. Then, the malicious packets injected into the system as FDI cyber-attacks were successfully detected and dropped by the developed AIS algorithm. Finally, the NARX intelligent network was used to successfully predict the lost/dropped packets. Therefore, the proposed method can be used by real-time ACPS to eliminate the effect of FDI cyber-attacks on aircraft, hence improving aircraft performances and flights' safety.

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